GORENSTEINNESS WITH LOCALLY NOETHERIAN FIBRES IS FPPF LOCAL ON THE TARGET AND SYNTOMIC LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0C0A (sources/stacks-duality/pinned-original-source.tex line 6844).

Allowed standard prerequisites (editorial).

- fppf locality of flatness on source and target. Descent, Lemmas labelled descent-lemma-descending-property-flat and descent-lemma-flat-fpqc-local-source, cited at proof lines 6856-6860. Disposes of the flatness part of the property.
- Local Noetherianity of a fibre after a finitely generated residue field extension. Varieties, Lemma labelled varieties-lemma-locally-Noetherian-base-change, cited at proof lines 6872-6876. Carries the local Noetherian hypothesis along an fppf cover.
- Base change of Gorensteinness. Same chapter, Lemma labelled lemma-gorenstein-base-change, cited at proof lines 6876-6878. Carries the Gorenstein property of the fibre to the base change.
- Syntomic locality of Gorensteinness of the fibres. Same chapter, Lemma labelled lemma-gorenstein-local-syntomic, cited at proof lines 6886-6889. Settles the source side.

Source reference. Original lemma 02L2: sources/stacks-chow/tags.txt, source lines 3393–3393. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6861 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 036K: sources/stacks-chow/tags.txt, source lines 4167–4167. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6862 of the counted proof. It is

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Source reference. Original lemma 038R: sources/stacks-chow/tags.txt, source lines 4246–4246. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6877 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0C03: sources/stacks-duality/pinned-original-source.tex, source lines 6480–6480. Same-chapter pointer into the author environment lemma at source line 6480; invoked at proof line 6880 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0C01: sources/stacks-duality/pinned-original-source.tex, source lines 6436–6436. Same-chapter pointer into the author environment lemma at source line 6436; invoked at proof line 6888 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"the fibres of f are locally Noetherian and f is Gorenstein" is local in the fppf topology on the target and local in the syntomic topology on the source.*

Proof. We have $\mathcal{P}(f) = \mathcal{P}_1(f) \wedge \mathcal{P}_2(f)$ where $\mathcal{P}_1(f) =$"f is flat", and $\mathcal{P}_2(f) =$"the fibres of f are locally Noetherian and Gorenstein". We know that $\mathcal{P}_1$ is local in the fppf topology on the source and the target, see Descent, Lemmas 02L2 and 036K. Thus we have to deal with $\mathcal{P}_2$.

Let $f : X \to Y$ be a morphism of schemes. Let $\{\varphi_i : Y_i \to Y\}_{i \in I}$ be an fppf covering of Y. Denote $f_i : X_i \to Y_i$ the base change of f by φ_i. Let $i \in I$ and let $y_i \in Y_i$ be a point. Set $y = \varphi_i(y_i)$. Note that

$$X_{i,y_i} = \mathrm{Spec}(\kappa(y_i)) \times_{\mathrm{Spec}(\kappa(y))} X_y.$$

and that $\kappa(y_i)/\kappa(y)$ is a finitely generated field extension. Hence if X_y is locally Noetherian, then X_{i,y_i} is locally Noetherian, see Varieties, Lemma 038R. And if in addition X_y is Gorenstein, then X_{i,y_i} is Gorenstein, see Lemma 0C03. Thus $\mathcal{P}_2$ is fppf local on the target.

Let $\{X_i \to X\}$ be a syntomic covering of X. Let $y \in Y$. In this case $\{X_{i,y} \to X_y\}$ is a syntomic covering of the fibre. Hence the locality of $\mathcal{P}_2$ for the syntomic topology on the source follows from Lemma 0C01. □